

Bodyweight Exercise based Exergame to Induce High Intensity Interval Training

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ABSTRACT

Exergames have been proposed as an attractive way of making exercise fun; however, most of them do not reach the recommended intensity. Although HCI research has explored how exergame can be designed to follow High Intensity Interval Training (HIIT) that is effective exercise consisting of intermittent vigorous activity and short rest or low-intensity exercise, there are limited studies on designing bodyweight exercise (BWE) based exergame to follow HIIT. In this paper, we propose BWE based exergame to encourage users to maintain high intensity exercise. Our initial study (n=10) showed that the exergame had a significant effect on enjoyment, while the ratio of incorrect posture (ex., squat) also increased due to participants' concentration on the exergame, which imply future design implications of BWE based exergames.

CCS CONCEPTS

• **Applied computing** → **Computer games**; • **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

Exergame, HIIT, Bodyweight Exercise

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1 INTRODUCTION

Although regular exercise is one of the most important factors in people's health, half of them who began the exercise stopped within six months [3]. The most common barriers for regular exercise are lack of time and motivation [7, 8]. Exergames, which combine exercise and games to enjoy exercising, have been proposed, but most exergames failed to meet the sufficient amount of exercise [6, 19]. To overcome the limitation, the exergames for high intensity exercise have been proposed [4, 10, 12].

HIIT is a particular example of time-effective exercise consisting of intermittent high-intensity exercise and low-intensity exercise

or rest [11, 20]. HIIT has been shown to have a more beneficial impact on fitness and health-related measures than traditional exercise in a shorter period of time [11]. Not only does it reduce the time needed for a healthy exercise regimen [11, 20], but it also provides people with enjoyment than low-to-moderation intensity exercise [5]. However, when doing HIIT training, it is a challenge to individuals in that they might lose their motivation at high intensity because hard exercise makes people feel exhausted and uncomfortable [16]. Therefore, several exergames incorporating HIIT have been proposed with an exercise machine, such as cycling or rowing [4, 10, 12], which use only one type of game input as cycling or rowing, but few studies have been conducted based on BWE movements consisting of various types.

In this paper, we propose a BWE-based rhythm exergame, inducing a player to perform training in time with the beat of the music in order to induce HIIT protocols. When body weight is used as the only resistance, the exercise can be performed without equipment [15]. It can also require a variety of movements from users, which may increase the fun of the exergame. Our study of 10 participants showed that the exergame increased their enjoyment above that of exercising by watching a YouTube-like video. However, the ratio of incorrect exercise posture increased due to the participants' concentration on the exergame. The results imply that the needs of game design that strongly induces accurate posture to avoid injury from incorrect posture (ex., bonus point to perfect posture and penalty to incorrect posture) in BWE-based exergame.

2 SYSTEM DESIGN

2.1 Bodyweight exercises based Exergame

While other exergames based on HIIT were usually designed to use an exercise machine, such as cycling or rowing with VR [4, 10, 12, 14], we use body weight exercise as game inputs because of its additional benefits. For example, body weight can be used as an adequate training load without the need for specialized facilities or equipment [15]. In addition, performing various body weight exercises (e.g., squat, lunge, jumping jacks, etc.) can be used as potential enjoyment points, compared to do one type of exercise using an exercise machine [9], and provides them with the opportunity to exercise all body parts. However, unlike cycling or rowing, in BWE, not only speed but also posture is an important factor. Therefore, the exergame should be designed to give sufficient time to perform exercise movements or to provide continuous feedback on the correct posture because some body weight exercises could pose a risk of injury due to wrong postures (e.g., squat) [21].

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Figure 1: Examples of the exergame screenshot

2.2 Exergame overview

Our exergame is a rhythm-based exergame utilizing the concept of performing body weight exercises to avoid hitting coming walls while synchronized with the music (Figure 1). In the usual rhythm games, while they give scores based on pressing the buttons on time, we calculate the score with two factors: not only (1) right timing; and but also (2) the accuracy of exercise posture. Since it is difficult to posture exactly on time, we provide a relatively low weight to the exact timing of performing the exercise, as well as high weight to exercise postures. Based on the two factors, whenever each time the users perform a dictated exercise to the beat of the music, the user gets a score of miss, good, or perfect. If users receive a good or a perfect score continuously, they will receive bonus points as a combo. The initial prototype was mostly focused on 'Squat' (and Jump), which is one of the most efficient and famous body weight exercises.

To provide a systematic overview, we describe the exergame along with MDA framework (standing for Mechanics, Dynamics, and Aesthetics) that is commonly used to conceptualize the dynamic behavior of game systems [13].

Mechanics: An avatar follows the movement of the player, whose goal is to avoid hitting upcoming walls that contains holes of various exercise shapes. In the initial design, there were only three exercises allowed including squat, holding squat and jump, which are similar motions and, as such, could be performed continuously to avoid tedious repetition. Unlike general rhythm game, there is only one note line, which allows the player to concentrate on the body weight exercise.

Dynamics: The gameplay follows a HIIT protocol with a number of high-intensity periods separated by rest periods. The number of high-intensity periods and the duration of each phase can be configured manually and depends on the length of music. During a high-intensity period, the walls advance toward the player in time to the beat of the music. The player is tasked with avoiding these walls by performing with the same exercise of the shape of the hole in the wall (Figure 1a, 1b, and 1c). During rest periods, the user can recover their stamina until the next high intensity period. During gameplay, a count for the running time, the number of exercises performed, and the score the user achieves are shown. Three seconds before a high intensity period starts, a message "HIGH-INTENSITY TIME!!!" is displayed in the center of the game screen, which is also shaken, to alert the user. The goal of the user is to perform the dictated exercise in time with the music (Figure 1d) so as to achieve a high score until the selected music is over.

Aesthetics: The overall feeling of the game is similar to other rhythm game [2]. The rest periods are aimed at evoking a restorative state by using a cyan palette (Figure 1e). When a high-intensity period is initiated, the screen changes into red color in order to evoke a sense of urgency.

3 IMPLEMENTATION

The exergame was implemented using Unity game engine with Kinect. The two major functional implementations are as follows.

- **Motion (Exercise) Tracking:** Kinect, a low-cost camera that can identify up to two moving players with 20 joints per person, was used to detect body weight exercise. By training exercise motion (e.g., squat) with Visual Gesture Builder, continuous body motion can be classified into squat, holding squat position, and jump. K2-Assets¹ was also used as a middleware to enable Kinect use in Unity.
- **Music Processing:** We manually divided the music into high intensity (about 20–30s) and rest period (about 10–15s) to match the music and to follow HIIT protocols [20]. In order to provide a note (timing for exercise motion) in time with the beat of music, Koreographer², an audio plugin for Unity, was used to synchronize music and gameplay.

4 EVALUATION

4.1 Participants

10 people participated in the initial study (1 female and 9 male, between the ages of 31 and 49, rated as low : 4, middle : 3, high : 3 in their self exercise level). They had all experienced squats before.

4.2 Study Procedure

We conducted within-subject lab study. Before the experiment started, the participants were asked to warm up for 10 minutes and conducted the two activities described below. To reduce order effects, the initial activity was counterbalanced. There was a brief pause for stamina recovery after the first activity. Each activity was conducted for only 2 minutes (Table 1) to avoid exhaustion.

- **Exercising with a video:** To compare the enjoyment of the proposed exergame to common exercise media as a baseline, we recorded an exercise video, as many people use a DVD/pre-recorded workout at home [1]. We did not use a common pre-recorded workout but instead recorded a new video with similar conditions to the exergame (e.g., # of

¹<https://assetstore.unity.com/packages/3d/characters/kinect-v2-examples-with-ms-sdk-and-nuitrack-sdk-18708>

²<https://assetstore.unity.com/packages/tools/audio/koreographer-professional-edition-44270>

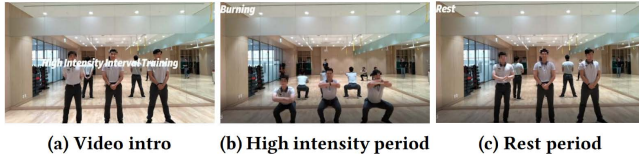


Figure 2: Recorded video examples

		1 SET	break	2 SET	break	3 SET
Video	# of squat	13		17		21
Exergame	# of squat	12		12		15
	# of hold	1		2		2
	# of jump	0		1		2
	duration	25s	13s	32s	16s	36s

Table 1: Experiment configuration for Video and Exergame

squat, exercise timing, and duration of each HIIT phase) for comparison purposes. In the video, three fitness experts we recruited did squats in time with the beat of the famous song, Psy's NAPAL BAJI [17] (Figure 2).

- Exergame: After a short tutorial of the exergame, the participants played the exergame using the same music as the video activity. The number and length of high-intensity and rest periods are exactly the same with the video activity, while only the exergame provided additional exercise motions (e.g., holding squat position and jump, described in Table 1) with different exercise timing compared to the video activity. Figure 3 shows a detailed exercise of 2 SET of both activities.

4.3 Data Collection

After finishing both sessions, the participants rated their enjoyment of the activities based on Intrinsic Motivation Inventory (IMI) [18] on a 7-point Likert scale, from 1 (not at all) to 7 (very true). After the questionnaire, we interviewed the participants. All activities were video-recorded for the posture analysis.

5 RESULTS

5.1 Enjoyment

A two-tailed paired t-test was performed which showed that the difference between the conditions for enjoyment were statistically significant. The questions and results are summarized in Table 2, and the p-values (< 0.05) were 0.001, 0.016, and 0.036, respectively.

We then analyzed the participants' interview following open-coding and axial-coding procedures to examine their responses. Regarding exercising by watching the video, all participants answered that the advantage was to learn correct exercise motion, while the disadvantage was that one-way interaction feels relatively boring. On the other hand, regarding the exergame, most participants felt it to be more fun. This was mostly due to the irregular rhythmic movement synched with the music, diverse exercise motion, and competition through scoring. However, they worried their incorrect posture because of insufficient exercise guidance and

	Questions	Video	Exergame
1	The activity was fun to do	3.6 (1.4)	5.8 (0.8)
2	I enjoyed doing the activity very much	3.9 (1.9)	5.8 (2.0)
3	While I was doing the activity, I was thinking about how much I enjoyed it	3.7 (3.0)	5.6 (2.5)

Table 2: Means (and variance) of the questionnaire for Video and Exergame. Answers are on a 7-point Likert scale ranging from 1 (not at all) to 7 (very true).

UserID	Video	Exergame
1	10%	32.3%
2	47.9%	71.7%
3	9.8%	28%
4	1.9%	10.7%
5	60%	71.4%
6	5.8%	57.6%
7	7.5%	6.4%
8	9.4%	44%
9	0%	6.2%
10	0%	28.8%

Table 3: The ratio of incorrect squat posture in the activity

their over-concentration on the game, forgetting the importance of exercise posture.

5.2 Squat posture accuracy

To quantitatively analyze the effect of the exergame on exercise posture, the squat posture of the participants during the two activities were analyzed based on the following rules³.

- (1) Push through the outer edges of the feet
- (2) Push the knees away from each other
- (3) Don't round or overarch the back

If any of the rules were not met, it was calculated as incorrect posture. The results are shown in Table 3 and a two-tailed paired t-test showed that the ratio of incorrect squat posture was significantly higher in the exergame group than the video group (p -value $0.0023 < 0.05$).

6 LIMITATIONS AND FUTURE WORK

We introduced an exergame for body weight exercise based on squat and similar actions. Our study showed that the exergame had a significant effect on enjoyment when compared to exercising by a watching video. When doing body weight exercise, correct posture is important to avoid a risk of injury. However, the ratio of incorrect posture was increased because of concentrating on the game in spite of that using posture accuracy for game scores. Therefore, a more robust strategy in consideration of this limitation is needed for further improvement.

REFERENCES

- [1] 2014. Global Consumer Exercise Trends Survey. Nielsen.
- [2] 2018. *Beat Saber*. Game [VR].

³<https://www.menshealth.com/fitness/g19547189/10-secrets-to-the-bodyweight-squat/?slide=1>

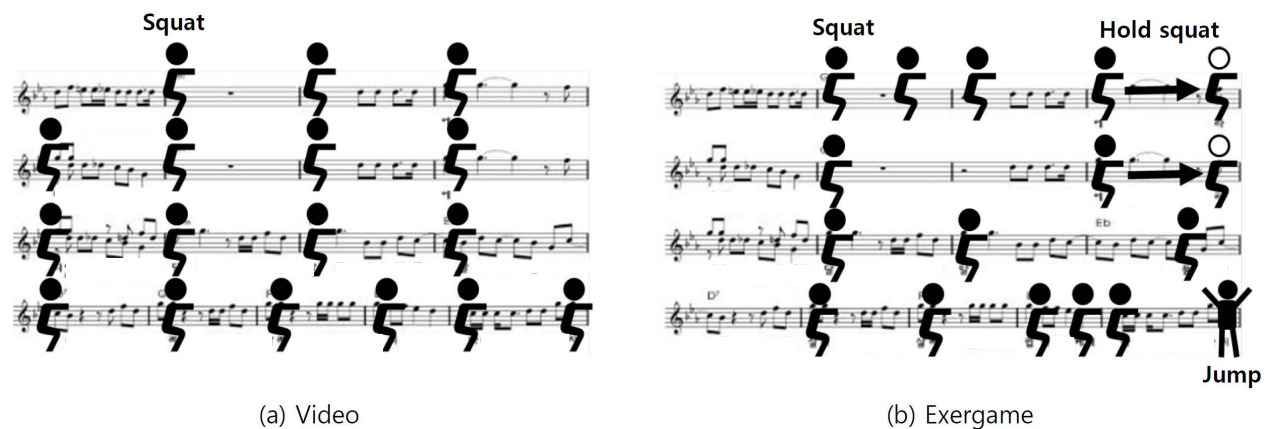


Figure 3: Detail example of 2 SET in the experiment. The position of the icon indicates the timing of the exercise.

- [3] James Annesi. 2003. Effects of a cognitive behavioral treatment package on exercise attendance and drop out in fitness centers. *European Journal of Sport Science* 3, 2 (2003), 1–16.
- [4] Soumya C Barathi, Daniel J Finnegan, Matthew Farrow, Alexander Whaley, Pippa Heath, Jude Buckley, Peter W Dowrick, Burkhard C Wünsche, James LJ Bilzon, Eamonn O'Neill, et al. 2018. Interactive feedforward for improving performance and maintaining intrinsic motivation in VR exergaming. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 408.
- [5] Jonathan D Bartlett, Graeme L Close, Don PM MacLaren, Warren Gregson, Barry Drust, and James P Morton. 2011. High-intensity interval running is perceived to be more enjoyable than moderate-intensity continuous exercise: implications for exercise adherence. *Journal of sports sciences* 29, 6 (2011), 547–553.
- [6] Elaine Biddiss and Jennifer Irwin. 2010. Active video games to promote physical activity in children and youth: a systematic review. *Archives of pediatrics & adolescent medicine* 164, 7 (2010), 664–672.
- [7] Michael L Booth, Adrian Bauman, Neville Owen, and Christopher J Gore. 1997. Physical activity preferences, preferred sources of assistance, and perceived barriers to increased activity among physically inactive Australians. *Preventive medicine* 26, 1 (1997), 131–137.
- [8] Ester Cerin, Evie Leslie, Takemi Sugiyama, and Neville Owen. 2010. Perceived barriers to leisure-time physical activity in adults: an ecological perspective. *Journal of physical activity and health* 7, 4 (2010), 451–459.
- [9] João Gustavo Claudino, Tim J Gabbett, Frank Bourgeois, Helton de Sá Souza, Rafael Chagas Miranda, Bruno Mezêncio, Rafael Soncin, Carlos Alberto Cardoso Filho, Martim Bottaro, Arnaldo Jose Hernandez, et al. 2018. Crossfit overview: systematic review and meta-analysis. *Sports medicine-open* 4, 1 (2018), 11.
- [10] Lucas Martins de Souza, Irem Gokce Yildirim, Anya Kolesnichenko, and Taiwoo Park. 2016. World of riders: exercising is fun. In *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts*. ACM, 55–60.
- [11] Martin J Gibala, Jonathan P Little, Maureen J MacDonald, and John A Hawley. 2012. Physiological adaptations to low-volume, high-intensity interval training in health and disease. *The Journal of physiology* 590, 5 (2012), 1077–1084.
- [12] Joshua C Haller, Young H Jang, Jack Haller, Lindsay Shaw, and Burkhard C Wünsche. 2019. HIIT The Road: Using Virtual Spectator Feedback in HIIT-based Exergaming. In *Proceedings of the Australasian Computer Science Week Multiconference*. ACM, 47.
- [13] Robin Hunicke, Marc LeBlanc, and Robert Zubek. 2004. MDA: A formal approach to game design and game research. In *Proceedings of the AAAI Workshop on Challenges in Game AI*, Vol. 4. 1722.
- [14] Aaron Keesing, Matthew Ooi, Ocean Wu, Xinghao Ye, Lindsay Shaw, and Burkhard C Wünsche. 2019. HIIT With Hits: Using Music and Gameplay to Induce HIIT in Exergames. In *Proceedings of the Australasian Computer Science Week Multiconference*. ACM, 36.
- [15] Brett Klika and Chris Jordan. 2013. High-intensity circuit training using body weight: Maximum results with minimal investment. *ACSM's Health & Fitness Journal* 17, 3 (2013), 8–13.
- [16] Wei Peng, Jih-Hsuan Lin, and Julia Crouse. 2011. Is playing exergames really exercising? A meta-analysis of energy expenditure in active video games. *Cyberpsychology, Behavior, and Social Networking* 14, 11 (2011), 681–688.
- [17] Psy. 2015. NAPAL BAJI. Video. Retrieved December 17, 2019 from https://www.youtube.com/watch?v=tF27TNC_4pc.
- [18] Richard M Ryan. 1982. Control and information in the intrapersonal sphere: An extension of cognitive evaluation theory. *Journal of personality and social psychology* 43, 3 (1982), 450.
- [19] G Stratton, N Ridgers, et al. 2008. Energy expenditure in adolescents playing new generation computer games. *Br. J. Sports Med* 42 (2008), 592–594.
- [20] Izumi Tabata, Kouichi Irisawa, MOTOKI Kouzaki, Kouji Nishimura, FUTOSHI Ogita, and Motohiko Miyachi. 1997. Metabolic profile of high intensity intermittent exercises. *Medicine and science in sports and exercise* 29, 3 (1997), 390–395.
- [21] Viviane Ugalde, Chuck Brockman, Zach Bailowitz, and Christine D Pollard. 2015. Single leg squat test and its relationship to dynamic knee valgus and injury risk screening. *PM&R* 7, 3 (2015), 229–235.